

Fundamental Structure (기본 구조)

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Linear Model

Response variable: Y , Explanatory variables: $x_1, x_2, \dots, x_p$

Model equation

$$E(Y) = \mu = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p$$

Assumptions (LINE)

Independence, homoscedasticity (equal variance), normality $Y_i \sim N(\mu_i, \sigma^2)$

Problem: What if the assumptions of homoscedasticity or normality are not satisfied?

Transform the data so that the above assumptions hold, and then apply the linear model

Or, propose a new model that relaxes or does not require these assumptions

Linear Model (Cont'd)

Example

Let $Y_i (i = 1, \dots, n)$ be independent and follow a Bernoulli distribution with success probability p_i

- (a) If $p_1 = \dots = p_n$ and $\theta_i = \log \frac{p_i}{1-p_i}$, then what are the likelihood, score function, and information matrix for θ ?
- (b) If we assume $\beta_0 + \beta_1 x_i = \log \frac{p_i}{1-p_i}$, then what are the MLE and information for β_0, β_1 ?

Notes $\frac{\partial l}{\partial \beta} = \sum_{i=1}^n \frac{\partial l_i}{\partial \theta_i} \frac{\partial \theta_i}{\partial \mu_i} \frac{\partial \mu_i}{\partial \eta_i} \frac{\partial \eta_i}{\partial \beta}, \eta_i = \beta_0 + \beta_1 x_i$

$\frac{\partial l_i}{\partial \theta_i}$: Relationship between likelihood and θ : exponential family (random component)

$\frac{\partial \theta_i}{\partial \mu_i}$: Relationship between θ and μ : exponential family (random component)

$\frac{\partial \mu_i}{\partial \eta_i}$: Relationship between μ and η : link function

$\frac{\partial \eta_i}{\partial \beta}$: Relationship between η and β : linear predictor (systematic component)

Generalized Linear Model

Model Equation

$$g(\mu) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

Assumptions

1. Independence: Observations are independent
2. Exponential Dispersion Family:

$$f(y; \theta) = \exp \left\{ \frac{y\theta - b(\theta)}{a(\psi)} \right\} + c(y, \psi)$$

- (i) The support $\{y: f(y) > 0\}$ does not depend on θ
- (ii) θ : Natural/canonical parameter (자연모수)
- (iii) ψ : Dispersion parameter (산포모수)

Generalized Linear Model (Cont'd)

Properties of the Exponential Family

$$E(Y) = \mu = b'(\theta), \text{ var}(Y) = b''(\theta)a(\psi)$$

Examples

Normal distribution: $f(y; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right)$

$$\theta = \mu, b(\theta) = \frac{\theta^2}{2}, a(\psi) = \sigma^2$$

$$\Rightarrow E(Y) = b'(\theta) = \theta = \mu, \text{ var}(Y) = b''(\theta)a(\psi) = \psi = \sigma^2$$

Poisson distribution: $f(y; \mu) = \frac{\mu^y e^{-\mu}}{y!}$

$$\theta = \log \mu, b(\theta) = \mu = e^\theta, a(\psi) = 1$$

$$\Rightarrow E(Y) = b'(\theta) = e^\theta = \mu, \text{ var}(Y) = b''(\theta)a(\psi) = e^\theta = \mu$$

Generalized Linear Model (Cont'd)

Bernoulli distribution: $f(y;\pi) = \pi^y (1 - \pi)^{1-y}$

$$\theta = \log \frac{\pi}{1 - \pi}, b(\theta) = \log(1 + e^\theta), a(\psi) = 1$$

$$\Rightarrow E(Y) = b'(\theta) = \frac{e^\theta}{1 + e^\theta} = \pi$$

$$\Rightarrow \text{var}(Y) = b''(\theta)a(\psi) = \frac{e^\theta}{(1 + e^\theta)^2} = \pi(1 - \pi)$$

Binomial distribution: $f(y;\pi) = \frac{n!}{y!(n-y)!} \pi^y (1 - \pi)^{n-y}$

$$\theta = \log \frac{\pi}{1 - \pi}, b(\theta) = n \log(1 + e^\theta), a(\psi) = 1$$

$$\Rightarrow E(Y) = b'(\theta) = n \frac{e^\theta}{1 + e^\theta} = n\pi \quad \Rightarrow \text{var}(Y) = b''(\theta)a(\psi) = n \frac{e^\theta}{(1 + e^\theta)^2} = n\pi(1 - \pi)$$

Generalized Linear Model Component

Random Component (랜덤 성분)

Distribution of the response variable $Y = (Y_1, \dots, Y_n)$: assumes independence of $Y_1, \dots, Y_n$

(Example) Counts of success - Binomial; Frequency - Poisson; Continuous - Normal

Systematic Component (체계적 성분)

A function of explanatory variables used to describe the mean of Y , $E(Y) = \mu$

Linear predictor (선형 예측): $\eta = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p$

Linear vs. Nonlinear

If the result after differentiation with respect to the parameter β does not include the parameter, the model is called linear

If it does, the model is called nonlinear

Generalized Linear Model Component (Cont'd)

Link Function (연결함수)

A function that connects the random component and the systematic component

For a monotone $g()$: $g(\mu) = \eta = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$

Example 1 - Identity function (항등함수)

When Y follows a normal distribution, $-\infty < \mu < \infty$:

$$\mu = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

Example 2 - logit function (로짓함수)

When Y follows a Bernoulli distribution, $0 < \mu < 1$:

$$\log \frac{\mu}{1 - \mu} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

When the logit function is used as the link function, the GLM is called a logistic regression model

Generalized Linear Model Component (Cont'd)

Example 3 - Log function (로그함수)

When $0 < \mu$:

$$\log \mu = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

Loglinear model (로그선형모형)

Generalized Linear Model Component (Cont'd)

Canonical link (정준연결)

When the natural parameter (θ) is used as the link function, i.e., $\theta = \eta$

(Example) Poisson Distribution

$$\begin{aligned} f(y; \mu) &= \exp[\{y \log(\mu) - \mu\} - \log(x!)] \propto \exp[y\theta - \exp(\theta)] \\ \Rightarrow g(\mu) &= \log(\mu) \end{aligned}$$

(Example) Binomial Distribution

$$\begin{aligned} f(y; \pi) &\propto \exp \left[y \log \left(\frac{\pi}{1-\pi} \right) + (n-y) \log(1-\pi) \right] \\ &= \exp[y\theta - (n-y) \log\{1 + \exp(\theta)\}] \\ \Rightarrow g(\pi) &= \log \left(\frac{\pi}{1-\pi} \right) : \text{logit function} \end{aligned}$$

Generalized Linear Model Component (Cont'd)

Properties of the link function

The sufficient statistic for θ is y

For a canonical link function, if $(x_i, y_i), i = 1, \dots, n$ are observed, then the sufficient statistic for β is xy

In the case of a canonical link function: $g(\mu) = (b')^{-1}(\mu)$ 이다.

Score function

$$\frac{\partial l}{\partial \beta} = \sum_{i=1}^n \frac{\partial l_i}{\partial \theta_i} \frac{\partial \theta_i}{\partial \mu_i} \frac{\partial \mu_i}{\partial \eta_i} \frac{\partial \eta_i}{\partial x_i} = \sum_{i=1}^n \frac{\partial l_i}{\partial \theta_i} \frac{\partial \eta_i}{\partial x_i}$$

(Example) Logistic regression coefficients β_0, β_1 can be obtained from the following estimating equations, given $(x_i, y_i), i = 1, \dots, n$:

$$(i) \quad 0 = \sum_{i=1}^n (y_i - b'(\theta_i))x_i = \sum_{i=1}^n \left(y_i - \frac{\exp(\eta_i)}{1 + \exp(\eta_i)} \right) x_i, \quad \eta_i = \beta_0 + \beta_1 x_i$$

$$(ii) \quad 0 = \sum_{i=1}^n (y_i - b'(\theta_i)) = \sum_{i=1}^n \left(y_i - \frac{\exp(\eta_i)}{1 + \exp(\eta_i)} \right), \quad \eta_i = \beta_0 + \beta_1 x_i$$

General Linear Model

General linear model (일반선형모형)

$$E(Y) = \mu = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

Regression model: explanatory variables = quantitative

ANOVA model: explanatory variables = qualitative

ANCOVA model: explanatory variables = quantitative+qualitative

Assumptions: independence, homoscedasticity, normality

If normality is assumed, the MLE of β = Least Squares Estimator (LSE)

Generalized linear model (일반화선형모형)

No need to transform data when analyzing non-normal data

Extension that includes not only the general linear model but also categorical data models

Parameter estimation is based on Maximum Likelihood Estimation (MLE)

Distributions belong to the exponential family

GLM for Binary Data

When the response variable has only two categories (success, failure):

$$P(Y = 1) = \pi, P(Y = 0) = 1 - \pi$$

Assume explanatory variables influence the probability of success

Linear probability model ($g(\mu) = \mu$)

Relationship between π and the linear predictor:

$$\pi(x) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

Structural issue

While $0 \leq \pi \leq 1$, the linear predictor satisfies $-\infty < \eta < \infty$

Thus, this formation can only be used meaningfully within limited range of x

If the estimated $\pi(x)$ exceeds 1 or falls below 0, errors occur (lack of convergence)

GLM for Binary Data (Cont'd)

Choice of Link Function

Problem and Solution

The response probability satisfies $0 \leq \pi \leq 1$, but the linear predictor $-\infty < \eta < \infty$

Tolerance distribution (허용분포)

For the distribution function $F(\cdot)$:

$$\mu = F(\eta) = \int_{-\infty}^{\eta} f(x)dx \rightarrow F^{-1}(\mu) = \eta$$

(Example) logistic distribution

$$\begin{aligned} f(x) &= \exp(x) / (1 + \exp(x))^2 \\ \rightarrow F(\eta) &= \exp(\eta) / (1 + \exp(\eta)) \end{aligned}$$

GLM for Binary Data (Cont'd)

Choice of Link Function

(Example) Normal distribution

$$f(x) = \frac{1}{\sqrt{2\pi}} \exp(-x^2/2)$$

$$\rightarrow F(\eta) = \Phi(\eta)$$

(Example) Extreme value distribution

$$f(x) = \exp[x - \exp(x)]$$

$$\rightarrow F(\eta) = 1 - \exp[-\exp(\eta)]$$

Logistic Regression Model

Logistic regression function (logit link function)

$$\log\left(\frac{\pi(x)}{1 - \pi(x)}\right) = \alpha + \beta x \quad \rightarrow \quad \frac{\pi(x)}{1 - \pi(x)} = \exp(\alpha + \beta x)$$
$$\rightarrow \pi(x) = \frac{\exp(\alpha + \beta x)}{1 + \exp(\alpha + \beta x)}$$

The inverse function of the logistic distribution is used as the link function

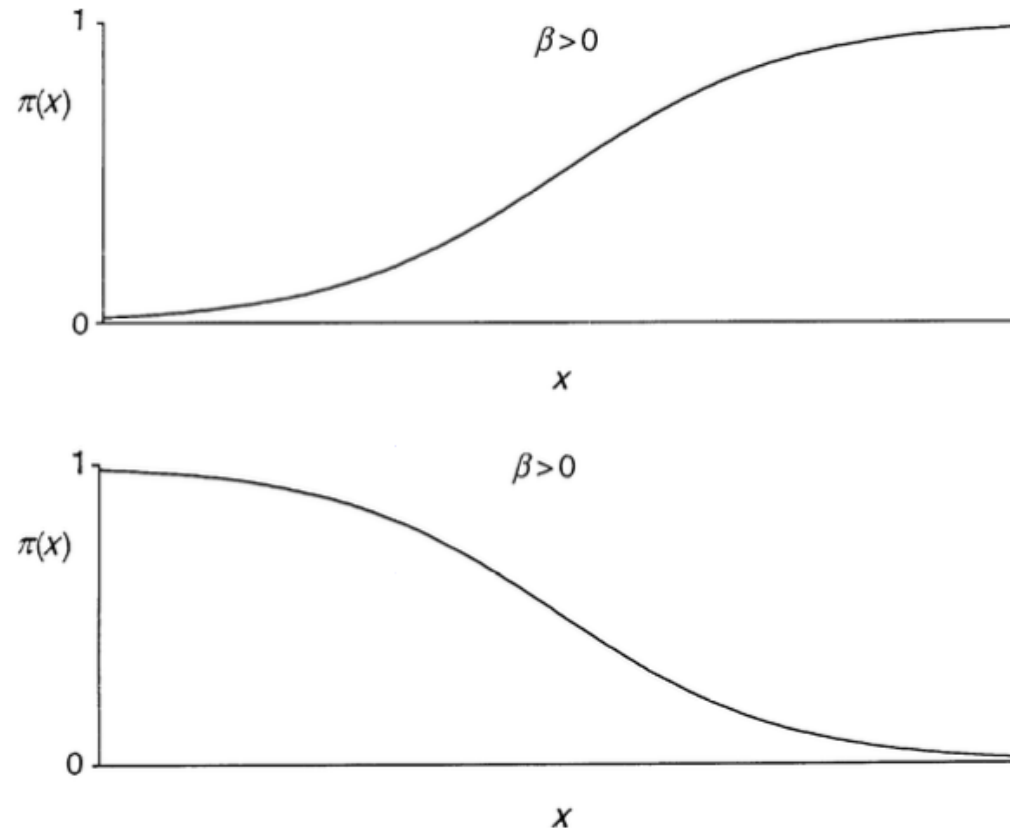
Uses the natural parameter as the canonical link function

If $\beta > 0$, then as x increases, $\pi(x)$ increases; if $\beta < 0$, it decreases

A larger $|\beta|$ indicates a greater increase (or decrease) in probability

Suitable for case-control studies because it allows estimation of the odds ratio

Logistic Regression Model (Cont'd)



<Logistic regression functions>

Logistic Regression Model (Cont'd)

Interpretation of Parameters

The effect of a one-unit increase in x :

Increases the log-odds by β

Increases the odds multiplicatively by $\exp(\beta)$

Slope of $\pi(x)$

$$\frac{\partial \pi(x)}{\partial x} = \frac{\beta e^{\alpha + \beta x}}{(1 + e^{\alpha + \beta x})^2} = \beta \pi(x)(1 - \pi(x))$$

When $\pi(x)$ is near 0.5, the effect of x is larger than when $\pi(x)$ is near 0 or 1

Logistic Regression Model (Cont'd)

Other Binary Links

Probit Model

Probit link function

Uses the CDF of the standard normal distribution as the link function

$$\pi(x) = \Phi(\alpha + \beta x) \rightarrow \Phi^{-1}(\pi(x)) = \alpha + \beta x$$

$\Phi(\alpha + \beta x)$ is the CDF of $(-\alpha + Z)/\beta$, where Z follows the standard normal distribution

When $\pi(x)$ is between 0.1 and 0.9, the probit link is nearly to the logit link

Complementary log-log model

$$\log[-\log(1 - \pi(x))] = \alpha + \beta x$$

Cloglog function

$$\rightarrow \pi(x) = 1 - \exp[-\exp(\alpha + \beta x)]$$

Uses the CDF of the extreme value distribution as the link function

Asymmetric model

Logistic Regression Model (Cont'd)

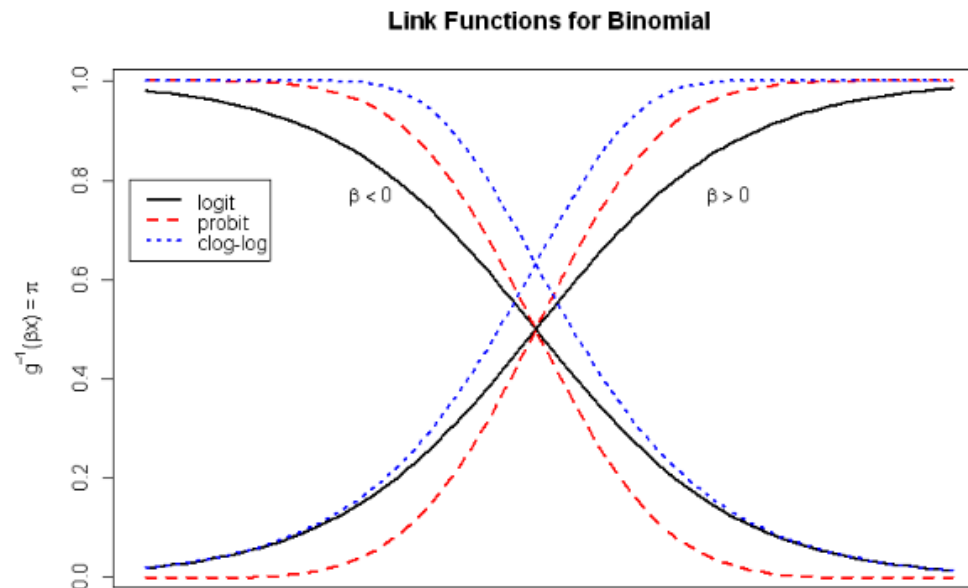
Other Binary Links

Log-log Model

loglog function

$$\begin{aligned} -\log [-\log (\pi (x))] &= \alpha + \beta x \\ \rightarrow \pi (x) &= \exp [-\exp (-\alpha - \beta x)] \end{aligned}$$

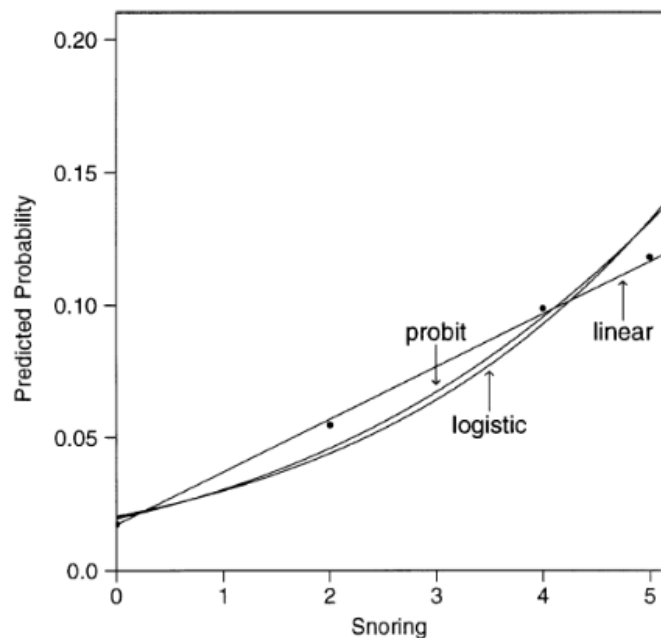
Asymmetric model



Logistic Regression Model (Cont'd)

(예제) Snoring and Heart Disease

코골이	Score	심장질환		비율	선형	로짓	프로빗
		예	아니오				
전혀 안함	0	24	1355	0.017	0.017	0.021	0.020
가끔씩	2	35	603	0.055	0.057	0.044	0.046
거의 매일	4	21	192	0.099	0.096	0.093	0.095
매일	5	30	224	0.118	0.116	0.132	0.131



Logistic Regression Model (Cont'd)

(예제) Beetle mortality after exposure to Carbon Disulfide

Log Dose	No. Bettles	No. Killed	Fitted Values		
			Cloglog	Probit	Logit
1.691	59	6	5.7	3.4	3.5
1.724	60	13	11.3	10.7	9.8
1.755	62	18	20.9	23.4	22.4
1.784	56	28	30.3	33.8	33.9
1.811	63	52	47.7	49.6	50.0
1.837	59	53	54.2	53.4	53.3
1.861	62	61	61.1	59.7	59.2
1.884	60	60	59.9	59.2	58.8

Logistic Regression Model (Cont'd)

Cloglog: $\hat{\alpha} = -39.52$, $\hat{\beta} = 22.01$

When $x = 1.7$, the survival probability $(1 - \hat{\pi})$ is

$$1 - \hat{\pi} = \exp\{-\exp(-39.52 + 22.01 \times 1.7)\} = 0.885$$

If x increases by 0.1, the survival probability $(1 - \hat{\pi})$ changes by $e^{22.01 \times 0.1} = 9.03$

That is, when log dose=1.8,

$$\exp\{-\exp(-39.52 + 22.01 \times 1.7 + 22.01 \times 0.1)\} = 0.885^{9.03} = 0.332$$

Goodness-of-fit statistic: $G^2 = 3.5 \sim \chi^2(df = 6)$

Logistic Regression Model (Cont'd)

Probit: $\hat{\alpha} = -34.89$, $\hat{\beta} = 19.74$

Estimation equation

$$\hat{\pi} = \Phi(-34.96 + 19.74x)$$

When $x = \frac{34.96}{19.73} = 1.77$, then $\hat{\pi} = 0.5$

$\hat{\pi}$ corresponds to the CDF of $N(\frac{34.96}{19.74}, (\frac{1}{19.74})^2)$

Goodness-of-fit statistic: $G^2 = 10.0 \sim \chi^2(df = 6)$